

◆ SECTION 3: Capacitance, Energy & Dielectrics (Q51–70)

51. Two capacitors, $C_1 = 4\mu F$ and $C_2 = 6\mu F$, are connected in series across 12 V. Voltage across C_1 :
- A) 4.8 V
 - B) 7.2 V
 - C) 6 V
 - D) 12 V
52. Two capacitors, $C_1 = 2\mu F$ and $C_2 = 3\mu F$, in parallel across 9 V. Charge on C_2 :
- A) 27 μC
 - B) 18 μC
 - C) 9 μC
 - D) 6 μC
53. Energy stored in 10 μF capacitor at 100 V:
- A) 0.05 J
 - B) 50 J
 - C) 5 J
 - D) 0.5 J
54. A 6 μF capacitor charged to 100 V is disconnected from battery. Dielectric with $K = 4$ inserted. New energy:
- A) $4 \times$ old
 - B) Old / 4
 - C) Same
 - D) $2 \times$ old
55. Capacitance of parallel plate capacitor with dielectric:
- A) $C = K\epsilon_0 A/d$
 - B) $C = \epsilon_0 A/Kd$
 - C) $C = \epsilon_0 A/d$
 - D) $C = K^2\epsilon_0 A/d$
56. Two identical capacitors in parallel combination store energy U. Series combination stores energy:
- A) $U/2$
 - B) $2U$
 - C) U
 - D) $4U$
57. Capacitor 4 μF connected across 200 V. Energy stored:
- A) 0.08 J
 - B) 80 mJ
 - C) 400 J
 - D) 0.8 J
58. Dielectric slab partially inserted in charged capacitor at constant voltage:
- A) Force pulls in slab
 - B) Force pushes out slab
 - C) No force
 - D) Energy decreases
59. Energy density in capacitor with dielectric:
- A) $\frac{1}{2}\epsilon E^2$

- B) ϵE^2
- C) $\frac{1}{2}E^2/\epsilon$
- D) E^2/ϵ

60. A 2 μF capacitor charged to 12 V is disconnected, dielectric $K=3$ inserted. Voltage across capacitor:
- A) 4 V
 - B) 12 V
 - C) 36 V
 - D) 6 V
61. Two capacitors $C_1=6 \mu\text{F}$, $C_2=3 \mu\text{F}$ in series across 12 V battery. Charge on C_2 :
- A) 12 μC
 - B) 24 μC
 - C) 6 μC
 - D) 18 μC
62. Capacitance of coaxial cable: inner radius a , outer radius b , length L :
- A) $2\pi\epsilon_0 L/\ln(b/a)$
 - B) $2\pi\epsilon_0 L\ln(b/a)$
 - C) $\epsilon_0 A/d$
 - D) $4\pi\epsilon_0 L/(b-a)$
63. Capacitor 10 μF charged to 100 V, connected to uncharged 10 μF capacitor. Final energy:
- A) 50% of initial
 - B) 75% of initial
 - C) Same
 - D) 25% of initial
64. Energy lost in above combination:
- A) 50%
 - B) 25%
 - C) 75%
 - D) 0
65. Dielectric inserted between plates at constant charge: energy:
- A) Increases
 - B) Decreases
 - C) Same
 - D) Zero
66. Capacitance of parallel plate, area 0.02 m^2 , separation 1 mm, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$:
- A) $1.77 \times 10^{-13} \text{ F}$
 - B) $1.77 \times 10^{-10} \text{ F}$
 - C) $1.77 \times 10^{-9} \text{ F}$
 - D) $1.77 \times 10^{-12} \text{ F}$
67. Energy stored in 2 μF capacitor at 100 V:
- A) 0.01 J
 - B) 10 J
 - C) 0.1 J
 - D) 1 J
68. For two capacitors in series, voltage divides:
- A) Inversely proportional to capacitance
 - B) Proportional to capacitance

- C) Equal
 - D) Depends on dielectric
69. Energy density in uniform field E between plates:
- A) $\frac{1}{2}\epsilon_0 E^2$
 - B) $\epsilon_0 E^2$
 - C) $E^2/2\epsilon_0$
 - D) Zero
70. Capacitor in parallel with battery, dielectric inserted:
- A) Charge increases
 - B) Charge decreases
 - C) Remains same
 - D) Zero

◆ SECTION 4: Gauss's Law & Electric Flux (Q71–90)

71. Electric field due to uniformly charged infinite plane sheet:
- A) $\sigma / 2\epsilon_0$
 - B) σ / ϵ_0
 - C) $2\sigma / \epsilon_0$
 - D) Zero
72. Electric flux through cube of side a enclosing charge Q :
- A) Q/ϵ_0
 - B) $6Q/\epsilon_0$
 - C) 0
 - D) $Q / 6\epsilon_0$
73. Flux through one face of above cube:
- A) $Q/6\epsilon_0$
 - B) Q/ϵ_0
 - C) $6Q/\epsilon_0$
 - D) 0
74. Electric field inside spherical conductor:
- A) Zero
 - B) Varies linearly
 - C) Constant non-zero
 - D) Maximum at surface
75. Electric field outside uniformly charged sphere (radius R , charge Q , $r > R$):
- A) $k Q / r^2$
 - B) $k Q / R^2$
 - C) Zero
 - D) Constant
76. Hollow spherical shell with charge Q : field inside cavity:
- A) Zero
 - B) $k Q / r^2$
 - C) Varies linearly
 - D) $k Q / R^2$

77. Flux through closed surface enclosing no charge:
A) Zero
B) Maximum
C) Minimum
D) Depends on shape
78. Line charge λ along infinite wire, flux through coaxial cylinder of radius r , length L :
A) $\lambda L / \epsilon_0$
B) $\lambda / 2\pi\epsilon_0 r$
C) $\lambda L / 2\epsilon_0$
D) Zero
79. Electric field near edge of finite sheet:
A) Uniform
B) Non-uniform
C) Zero
D) Infinite
80. Gauss's law used to find field of:
A) Infinite plane sheet
B) Spherical charge
C) Infinite line charge
D) All above
81. Field inside non-conducting solid sphere with uniform charge density at distance r :
A) $k Q r / R^3$
B) $k Q / r^2$
C) Zero
D) $k Q / R^2$
82. Electric flux through surface perpendicular to uniform E :
A) $E \cdot A$
B) $E \cdot A \cdot \cos\theta$
C) Zero
D) $E \cdot A \cdot \sin\theta$
83. Electric field just outside charged conductor:
A) σ / ϵ_0
B) $2\sigma / \epsilon_0$
C) Zero
D) $\sigma / 2\epsilon_0$
84. Two coaxial spherical shells, inner Q , outer uncharged. Field between shells:
A) $k Q / r^2$
B) Zero
C) Varies linearly
D) Depends on shell thickness
85. Potential inside conducting sphere:
A) Constant
B) Zero
C) Varies linearly
D) Maximum at center
86. Gauss surface inside a conductor in electrostatic equilibrium:
A) Field = 0
B) Field = maximum

- C) Field varies linearly
 - D) Flux non-zero
87. Electric field at surface of charged spherical shell:
- A) $k Q / R^2$
 - B) Zero
 - C) $k Q / r^2$
 - D) Varies
88. Field inside cylindrical conductor:
- A) Zero
 - B) Varies linearly
 - C) Constant
 - D) Maximum
89. Electric flux through open surface in uniform E:
- A) $E \cdot A \cdot \cos\theta$
 - B) $E \cdot A \cdot \sin\theta$
 - C) Zero
 - D) $E \cdot A$
90. Electric field due to thin spherical shell outside shell:
- A) Like point charge
 - B) Zero
 - C) Constant
 - D) Varies linearly